AC VOLTAGE CONTROLLER

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AC voltage controllers (regulators) changes the magnitude of ac voltage without change of frequency

Types :

a) Phase angle control – Switches connect the load to the ac source for a portion of each cycle of input voltage.

b) Integral cycle control – Switch connect the load to the source for an integral number of cycles of input voltage and disconnect it for another integral number of cycles.

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Types :

c) Sequence Control – Two or more stages of voltage controllers are used in parallel; they are triggered in proper sequence one after the other so as to obtain a variable output with low harmonic content

Applications : Industrial heating, induction heating, speed control of induction motors, static reactive power control, lighting control etc. in anti-parallel.

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PHASE ANGLE CONTROL

- Switches connect the load to the ac source for a portion of each cycle of input voltage.
- Thyristors are line commutated;
- No need of complex commutation circuit.

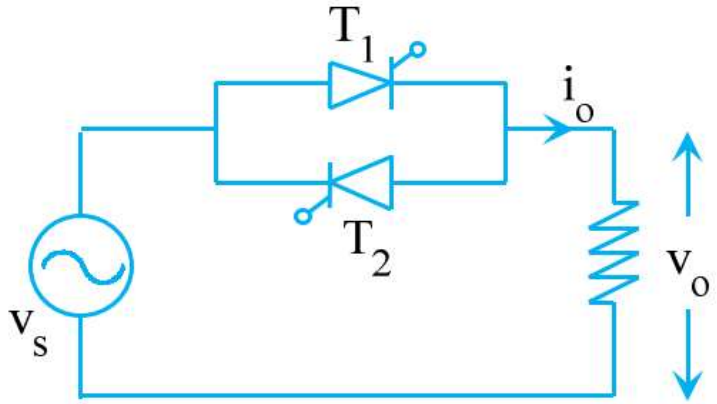
Disadvantage:

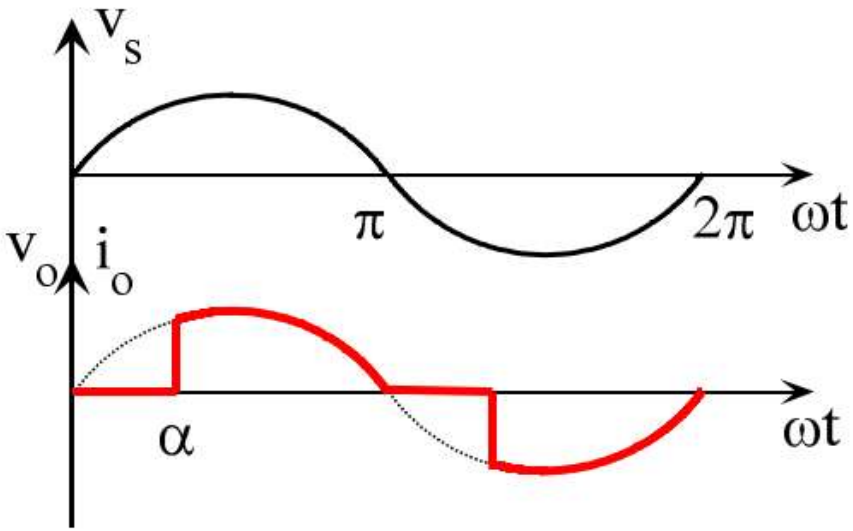
- Presence of harmonics in the supply current and load voltage

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1-Phase Full Wave AC voltage controller

- Two thyristors are used in anti-parallel.





Here, positive and negative areas of v_o are same; hence no dc component.

RMS value of output voltage, $V_o = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi} V_m^{\,2} \sin^2 \omega t d\omega t} = V_m \sqrt{\frac{1}{2\pi}(\pi - \alpha + \frac{\sin 2\alpha}{2})}$

When $\alpha = 0$, $V_o = \frac{V_m}{\sqrt{2}} = V_s$ and when $\alpha = \pi$, $V_o = 0$.

By varying $\alpha = 0$ to π, RMS value of output voltage V_o can be controlled from V_s to 0.

RMS value of load current, $I_o = \frac{V_o}{R}$

Input power factor = Real power/Apparent power

Apparent power = Input VA = $V_s I_s = V_s I_o = \frac{V_s V_o}{R}$

Output (real) power = $\frac{V_o^2}{R}$

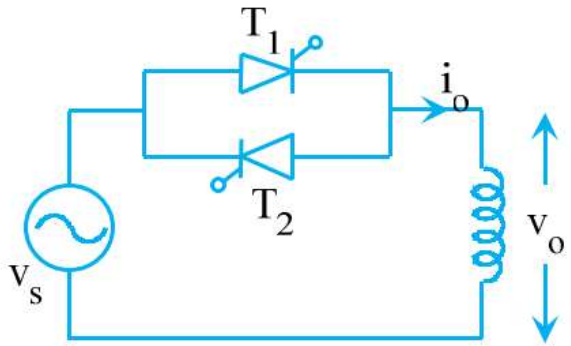
Input power factor = $\frac{V_o}{V_s} = \sqrt{\frac{1}{\pi}(\pi - \alpha + \frac{\sin 2\alpha}{2})}$

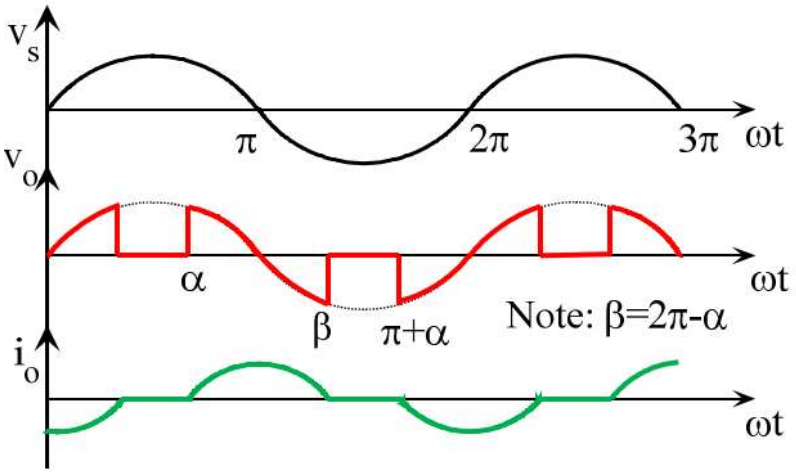
Problem : A 1-phase full wave ac voltage controller has a resistive load of 20Ω while the input voltage is 220V at 50Hz. If the firing angles of each of the anti-parallel thyristors are 90°, calculate a) RMS value of the output voltage, b) RMS value of the input voltage and c) the input power factor.

Solution : $V_o = V_m\sqrt{\frac{1}{2\pi}(\pi - \alpha + \frac{\sin 2\alpha}{2})} = \sqrt{2}\times 220\times\sqrt{\frac{1}{2\pi}(\pi - \frac{\pi}{2} + \frac{\sin 2\times\pi/2}{2})}$ =155.56V

$I_o = \frac{V_o}{R}$ =7.78A; Input power factor = $\frac{V_o}{V_s}$ =0.707 (lag)

AC VOLTAGE CONTROLLER with L LOAD





During positive half cycle, T_1 is triggered at $\alpha \geq \pi/2$. When T_1 is conducting, $v_o=v_s$. During negative half cycle, T_1 will continue to conduct upto β (β =extinction angle = $2\pi-\alpha$) due to stored energy in the inductance. (Note : Average voltage across inductance is zero; equal positive & negative areas for voltage). T_2 is triggered at $\pi+\alpha$. When T_2 is conducting, $v_o=v_s$. T_2 conducts upto $3\pi-\alpha$.

For a voltage controller with L load,

$$L\frac{di}{dt} = V_m \sin \omega t \text{ where } \alpha \le \omega t \le \beta$$

$$i = \frac{1}{L}\int V_m \sin \omega t d\omega t$$

$$= -\frac{V_m}{\omega L}\cos \omega t + K$$

At $\omega t = \alpha,\ i = 0$

Hence, $-\frac{V_m}{\omega L}\cos \alpha + K = 0$

$$K = \frac{V_m}{\omega L}\cos \alpha$$

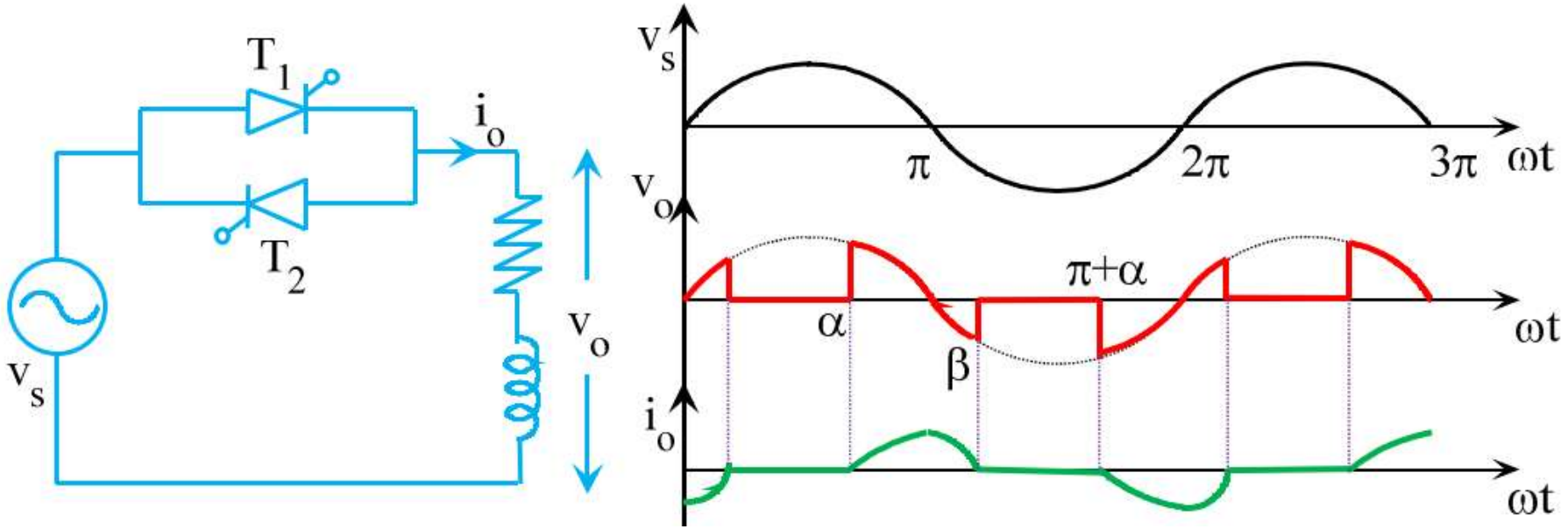
$$i_o(t) = \frac{V_m}{\omega L}(\cos \alpha - \cos \omega t)$$

If $\alpha = \pi / 2$,

$$i_o(t) = -\frac{V_m}{\omega L}\cos \omega t = \frac{V_m}{\omega L}(\sin \omega t - \pi / 2)$$

If $\alpha = \pi / 2$, output voltage and input voltage waveforms are same. Load is directly connected to ac source.
Hence, for L load, output voltage can be controlled only if $\alpha \geq \pi / 2$.

AC VOLTAGE CONTROLLER with RL LOAD



During positive half cycle, at $\omega t = \alpha$, T_1 is triggered. When T_1 is conducting, $v_o = v_s$. During negative half cycle, T_1 will continue to conduct upto β (where β = extinction angle; $\beta - \pi \approx \Phi = \tan^{-1}\frac{\omega L}{R}$) due to stored energy in the inductance. T_2 is triggered at $\pi+\alpha$. When T_2 is conducting, $v_o = v_s$. T_2 conducts upto $\pi+\beta$.

If $\alpha \leq \Phi$, circuit behaves as if RL load is directly connected to ac source.

APPLICATIONS OF AC voltage controllers are

- Heating: Regulating temperature in both domestic and industrial heating devices.
- Lighting: Controlling the brightness of lights, often referred to as dimming.

- Speed control: Adjusting the speed of single-phase and three-phase AC motors, commonly found in fans, pumps, and other drives.
- Induction motor starting: Used to control the starting process of induction motors.
- Transformer tap changing: Adjusting the voltage output of a transformer while it is still connected to the load (on-load tap changing).
- AC magnet control: Controlling the strength of AC magnets.